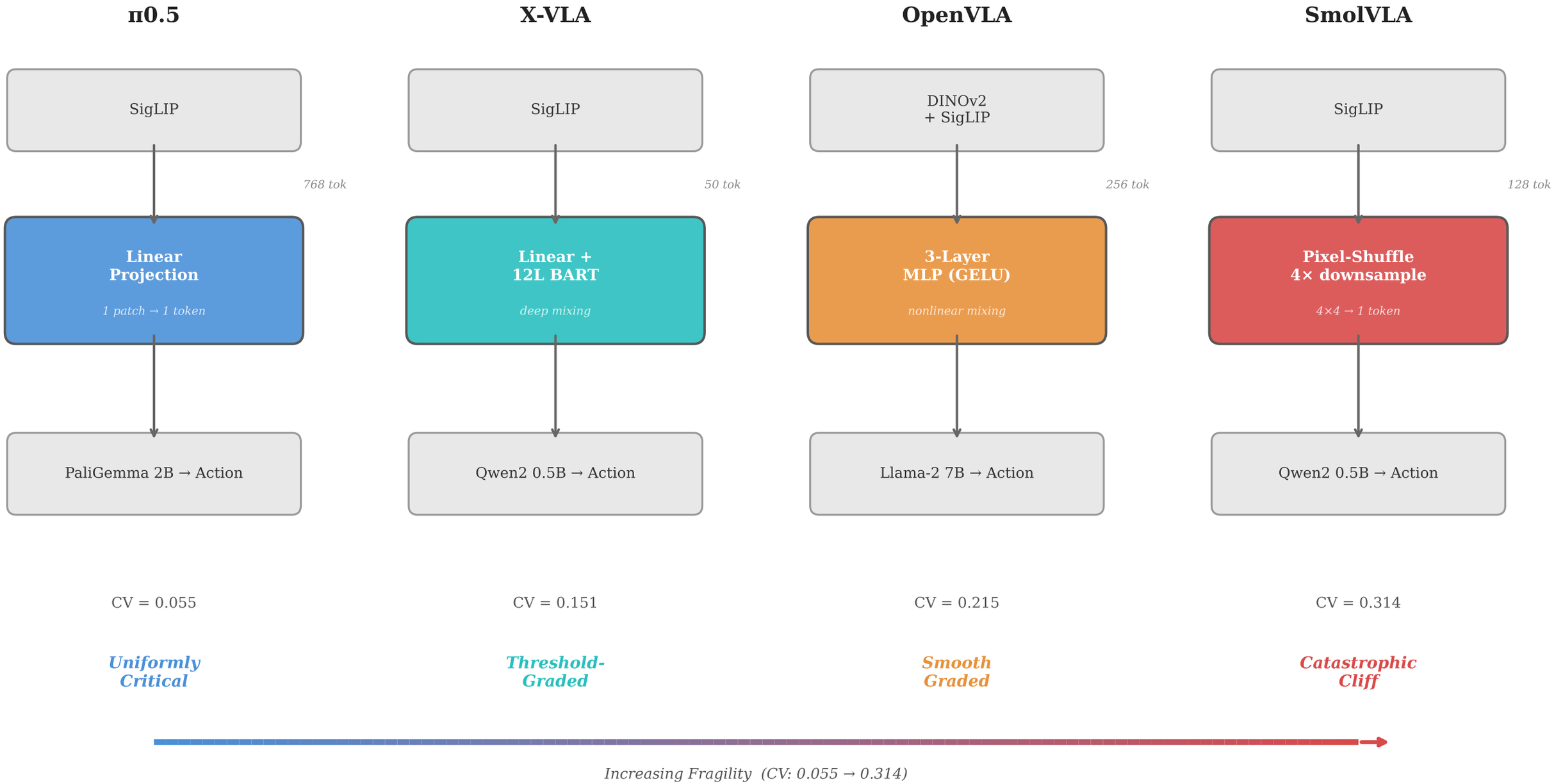

π0.5
X-VLA
OpenVLA
SmolVLA
SigLIP
SigLIP
DINOv2
+ SigLIP
SigLIP
768 tok
50 tok
256 tok
128 tok
Linear
Projection
1 patch → 1 token
Linear +
12L BART
deep mixing
3-Layer
MLP (GELU)
nonlinear mixing
Pixel-Shuffle
4× downsample
4×4 → 1 token
PaliGemma 2B → Action
Qwen2 0.5B → Action
Llama-2 7B → Action
Qwen2 0.5B → Action
CV = 0.055
CV = 0.151
CV = 0.215
CV = 0.314
Uniformly
Critical
Threshold-
Graded
Smooth
Graded
Catastrophic
Cliff
Increasing Fragility (CV: 0.055 → 0.314)